THE RELATIONSHIP BETWEEN STIMULUS PARAMETERS AND PHOSPHENE THRESHOLD/BRIGHTNESS, DURING STIMULATION OF HUMAN VISUAL CORTEX

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Electrical stimulation of the surface of human visual cortex evokes small sensations of light[1,2], called "phosphenes", which offer a potential means for developing a visual prosthesis for the blind[3]. Such a prosthesis would bypass the normal retinocortical connections and, employing an external photo-imaging device, transmit images directly to the cortex. Transmission of very simple static and changing patterns in this manner has already been accomplished[4].

Development of the technology for cortical stimulation in humans using chronically implanted electrodes[5] makes it possible to chart the input/output characteristics of what is an essentially unexplored sensory system. Construction of such a psychophysical profile is obviously essential to development of a prosthesis, and it may generalize to some extent to stimulation of other cortical areas. Recent experiments indicate that all the empirically-determined psychophysical relationships are predictable on the basis of a few fundamental characteristics of the participating neural elements.

Though they represent considerable over-simplification, the following algorithms can be used to predict stimulus-response relationships with a tolerable amount of error:

1) For trains of stimulus pulses at any specified set of parametric values, the sensory response will be determined by only those pulses delivered in the first 100 msec.

2) Within this 100 msec, there occurs complete summation of pulse duration over time. One 1.0 msec pulse has the same effect as two 0.5 msec pulses or four 0.25 msec pulses.

3) The amount of current required to produce a given sensory effect, whether threshold or a fixed supra-threshold brightness, is related to cumulative pulse duration, or "pulse time", according to an inverse power function.

4) For a given value of pulse time, the magnitude or brightness of the sensation is logarithmically related to the stimulating current.

METHODS

Subject and Hardware. These experiments have been conducted with a blind volunteer subject who underwent surgical implantation about 4 yrs ago of a 64-electrode array. The array rests in contact with the medial surface of the right occipital lobe. The platinum electrodes are arranged in 8 rows of 8, hexagonally positioned in a teflon paddle so that inter-electrode spacing is a uniform 3 mm center to center. Sixty-four insulated wires travel subcutaneously from the array to an electrical connector which protrudes through the scalp behind the ear. Stimulation is controlled by a special computer system[5]. More detailed accounts of the methods and results of the experiments presented here are being prepared for publication elsewhere.

Phosphene Characteristics and the Stimulus. The phosphenes evoked by stimulation with small (1 mm diameter) electrodes are generally punctate (diameter about 1° visual angle) colorless sensations. Visual field location is stable and dependent solely on the cortical locus of stimulation. Stimulation is in the form of repetitive biphasic rectangular constant-current pulses and is specifiable in terms of current I, frequency F, train duration T, biphasic pulse duration t_p, and number of pulses P. Manipulation of these waveform variables generally affects only the brightness of the phosphene. Experiments on basic psychophysics show that current required for threshold decreases in various fashions with increases in any of the other variables, while brightness grows with increases in each variable including I.

RESULTS

Threshold Sensations

Figure 1 presents results of a threshold experiment in which the parameters manipulated were number of pulses P, pulse duration t_p, and train duration T. (Varying the train duration of a fixed number of pulses is in effect varying the frequency at which they are delivered.) The data reflect 2 kinds of temporal integrative processes: summation of pulses over train time and summation of current over pulse time.

Train-Time Summation. For any given number of pulses at any given pulse duration in Figure 1, the associated threshold function is essentially flat across train durations. For example, threshold for two 0.25 msec pulses remains roughly constant as the interval between those 2 pulses is increased from 5 to 20 msec. Moreover, this 2-pulse threshold value is equal to that of a single 0.50 msec pulse. The system, then, may be said to

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be performing complete temporal integration, over train time, of the effects of the individual pulses; threshold is consequently independent of train duration (within the range of T-values used here).

Pulse-Time Summation. The variables pulse duration and number of pulses each exert significant leverage over threshold, determining the ordinate location of each flat function. In fact, given the equivalency of two 0.25 msec pulses and a single 0.50 msec pulse, these data suggest that the critical factor in determining threshold current is the product of P and t_p, or cumulative pulse duration. This quantity, which we term "pulse time" and denote by the symbol t, is in effect the individual pulse durations collapsed over train time. As Figure 1 shows, the larger the pulse time value associated with the stimulus, the lower its threshold.

The Utilization Function. The nature and degree of pulse time summation are demonstrated more directly in Figure 1B. The data points of Figure 1A are here replotted as a function of pulse time. In theory, each flat function of Figure 1A collapses onto a single point. The solid line fit to the data is termed the system's "utilization function". This function is linear in logarithmic coordinates and implies an inverse power function:

$$I = Kt^{-c} \quad \text{or equivalently,} \quad It^{c} = K \qquad \text{Equation 1}$$

The values of the intercept constant K and slope (absolute value) constant c vary somewhat among electrodes. The least squares fit for the 3 electrodes represented in Figure 1 yields $I = 2t^{-0.4}$.

The utilization function expresses a tradeoff between the amount of current required for threshold and the cumulative pulse duration of the stimulus. In Figure 1A, 2 of the points on the utilization function are single-pulse thresholds. In fact, the simplest conceptualization of this function is that of a current-duration tradeoff for a single-pulse stimulus at threshold. Because of train-time integration, as Figure 1A demonstrates, multiple-pulse thresholds collapse onto the same utilization function as do single-pulse thresholds. Train-time summation of pulses, then, is complete, while pulse time summation, because the relationship between I and t is not reciprocal, is incomplete. The relationship expressed by Equation 1 reflects instead a degree of partial integration of current over pulse time.

The Critical Duration. Obviously, there must be a limit to the train-time duration over which successive pulse will be completely integrated. Two different perspectives of this "critical duration" are embodied in the data presented in Figure 1.

One set of data points represents results of a threshold experiment in which the stimulus was always a train of four 0.5 msec pulses; the independent variable was the train duration over which these 4 pulses were distributed. This paradigm then is analogous to that which applies to any given function in Figure 1A but it extends the range of train durations used. Since the pulse-time value associated with the 4-pulse train was always the same, 2 msec, threshold should remain the same as long as train-time summation is complete. The data function is essentially flat for train durations up to about 60 msec, whereafter threshold begins to rise progressively. It may be said, then, that train-time summation begins to fail at about 60 msec. We do not, however, assign this value to the critical duration, since the breakdown of integration is not abrupt but rather occurs gradually over a train duration interval from about 60 msec to about 175 msec. Instead, we define critical duration T_c as the logarithmic midpoint of this transition interval and use the value 100 msec as a rough approximation. Moreover, for purposes of creating a mathematical model, the system is portrayed as if the transition from complete pulse integration to complete pulse independence occurred at 100 msec abruptly.

Threshold rises with increasing train duration as successively fewer of the 4 pulses occur within the first 100 msec of the train. For the last 2 data points on this function, the number of pulses occurring within T_c is respectively 3 and 2. When train duration reaches a value (which it does not in the data of Figure 1) for which the 4 pulses are so widely distributed in time that no temporal integration occurs among them - that is, for which the interpulse interval significantly exceeds the critical duration - the threshold should again become constant, at a value equal to threshold of a single pulse.

The second set of data points in Figure 2 provides a different view of the same integrative processes. In this experiment, train duration was varied in the more usual way by increasing the number of pulses delivered at a fixed frequency (here, 50 pps.). Because increasing the train duration produces an increase in the amount of pulse time, it also produces a decrease in threshold until T exceeds T_c and train-time summation fails completely. This procedure, then, is sensitive to the upper limit of the T_c transition interval, beyond which the later pulses cause no further drop in threshold and are therefore functionally redundant.

Suprathreshold Sensations

Constant Brightness. Brightness-matching experiments, in which the subject adjusts the current over trials of a test phosphene flash so that its brightness is equal to that of a constant nearby phosphene, produce results that are analogous to those obtained for threshold (which is in effect a constant, near-zero brightness). For a fixed number of pulses and fixed (short) train duration, variations in pulse duration yield the same kind of tradeoff between current and pulse time as that which applies at threshold. In fact the constant-brightness

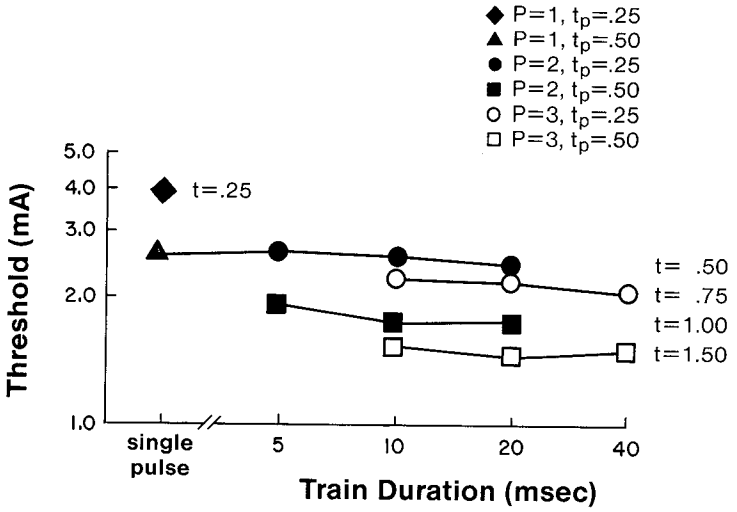

Figure 1A. Threshold current as a function of train duration with pulse duration t_p and number of pulses P as parameters. The value indicated to the right of each function is the associated pulse time t, where $t = Pt_p$.

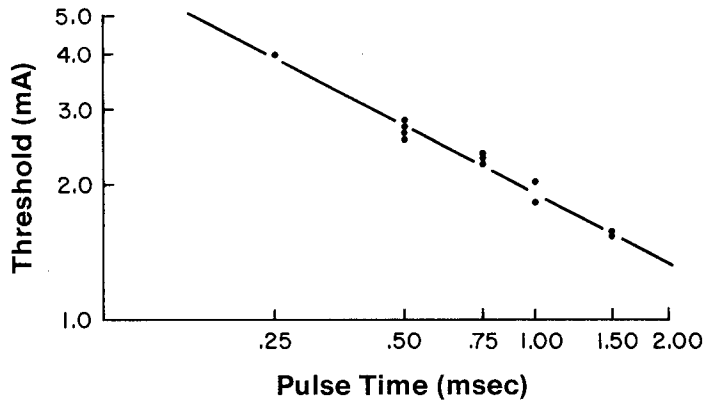

Figure 1B. Utilization function: threshold current as a function of pulse time. Data points are the same as in Figure 1A.

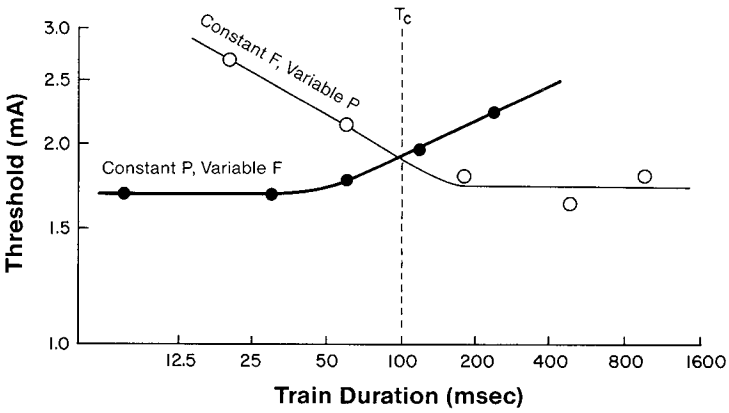

Figure 2. Threshold current as a function of train duration: (0) four 0.5 msec pulses, (0) variable number of pulses (2, 4, 10, 25, and 50) at a frequency of 50 pps.

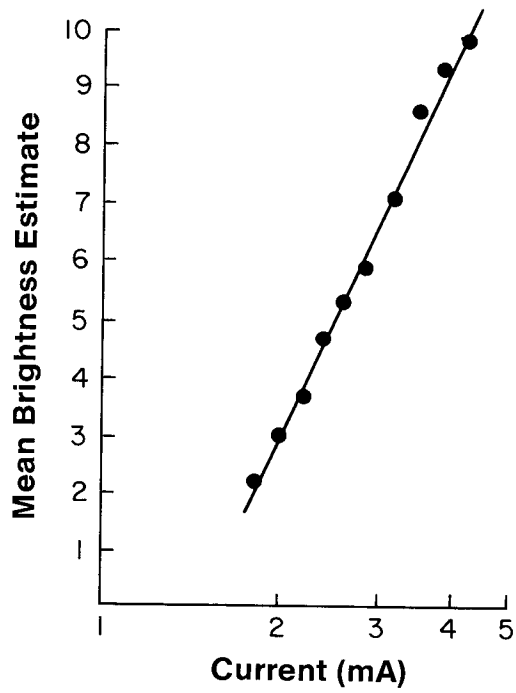

Figure 3. Mean brightness estimate as a function of current (one electrode). Stimulus parameters were: T = 500 msec, F = 50 pps, t_p = 0.5 msec.

utilization functions appear to parallel the threshold function; functions for higher brightness are located higher on the ordinate. Also, train-time summation for suprathreshold sensations occurs up to a T_c of about 100 msec as it does for threshold sensations.

Brightness as a Variable. Figure 3 presents results of a brightness magnitude estimation experiment in which current I was the independent variable and temporal variables were constant (T = 500 msec, F = 50 pps, t_p = 0.5 msec). The data are linear in semilogarithmic coordinates and reflect the relationship

$$B = K_1 \log I - K_2 \qquad \text{Equation 2}$$

where B is brightness and K_1 and K_2 are constants. Other experiments show that at a given current, increasing the train duration beyond 100 msec does not increase the stimulus brightness, so that the same B vs I function which applies at the critical duration also describes data for longer trains.

DISCUSSION

For purposes of predicting input/output functions, probably the most important finding reported here is the complete summation of successive pulses over train time. This means that all of the variables which temporally distribute the stimulus energy - i.e., T, F, P and t_p - exercise no direct leverage on the induced sensory effect but rather influence it only by influencing the amount of pulse time occurring within the critical duration. We term this latter quantity effective pulse time t_c and generally evaluate it at T = 100 msec.

The relationships established here can be represented mathematically by a single equation which embodies Equations 1 and 2:

$$B = C_1 \log I + C_2 \log t_c - C_3 \qquad \text{Equation 3}$$

where the constants are derived from those of Equations 1 and 2 and t_c is the amount of pulse time occurring in the first 100 msec of stimulation.

Since pulse time may be defined mathematically in terms of combinations of the standard variables F, T, P, and t_p, Equation 3 may be used to generate theoretical versions of all the psychophysical functions obtained empirically. For threshold functions, B is assigned a value of zero, and Equation 3 reduces to Equation 1 (the exponential constant c is contained in C_2). For magnitude-intensity functions, temporal parameters are constant and Equation 3 reduces to Equation 2.

For the objective of developing a prosthetic device, the present findings have some important implications. First, they provide a fairly simple code for programming the transformation of the electrical stimulus into a known sensory response at each electrode contributing to the composite half-tone image. Second, the finding that successive pulses are temporally integrated over any 100 msec interval implies that the "frame" rate for information transmission must not be greater than 10 per second. Third, the present findings provide some rules for minimizing several charge-related stimulus measures, with the aim of minimizing chances of adverse physiological effects of long-term stimulation. For example, the utilization function shows that the charge (as well as the power requirements) necessary to induce a given sensory effect is least for stimuli with short pulse times. On the other hand, current and current density are minimized by using stimuli with long pulse times. Moreover, the rate (over train time) of delivery of all these charge-related quantities can be minimized by parcelling out the pulse time in temporally distribution pulses within the critical duration.

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DISCUSSION OF MANUSCRIPT #71

DR. WOODARD: How is the curve of current versus threshold stimulation affected by electrode size?

DR. HENDERSON: In general, electrode size doesn't seem to affect the threshold. Fifty-two of the electrodes in this patient are one mm in diameter, and twelve are 2 mm in diameter in the 64 electrode array. We have found no systematic differences in psychophysical results.

DR. KANTROWITZ: Any chance for a prosthesis that can fit down in the calcarine sulcus? This, I presume, is way out in the peripheral field.

DR. HENDERSON: It's unlikely in the foreseeable future since the calcarine fissure is so variable in size and extent, and very thin flexible electrodes would be difficult to fabricate.

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